1 Black Holes with a Negative Mass

(a) The Schwarzschild metric is

$$ds^2 = -\left(1 - \frac{2G_N M}{r}\right) dt^2 + \left(1 - \frac{2G_N M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \tag{1}$$

from which we see that changing r to $-r$ is equivalent to changing M to $-M$ because in both cases the metric becomes

$$ds^2 = -\left(1 + \frac{2G_N M}{r}\right) dt^2 + \left(1 + \frac{2G_N M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \tag{2}$$

Equation (2) is the Schwarzschild metric with a negative mass, but since we have explicitly written out the minus sign of the mass, in Eq. (2) we have $M > 0$ and $r > 0$.

(b) $r = 0$ is still a genuine singularity. In the lecture notes, we showed that for the Schwarzschild metric, the Kretschmann scalar

$$K = \frac{48 G_N^2 M^2}{r^6} \tag{3}$$

blows up at $r = 0$, so we concluded that $r = 0$ is a genuine singularity for the Schwarzschild metric. In the metric Eq. (2), the only difference from the Schwarzschild metric is that we have changed M to $-M$, but the Kretschmann scalar in Eq. (3) is proportional to M^2, so the Kretschmann scalar is unchanged in the new metric. We therefore conclude that $r = 0$ is still a genuine singularity.

(c) Examining the new metric, we find that $-\left(1 + \frac{2G_N M}{r}\right)$ is always smaller than zero. Therefore, t is everywhere timelike.

(d) Similarly, $\left(1 + \frac{2G_N M}{r}\right)^{-1}$ is always larger than zero, so r is everywhere spacelike and never timelike.

(e) For a radial null trajectory, Eq. (2) becomes

$$0 = -\left(1 + \frac{2G_N M}{r}\right) dt^2 + \left(1 + \frac{2G_N M}{r}\right)^{-1} dr^2, \tag{4}$$

from which we get (for an outward trajectory where r increases with t)

$$\frac{dt}{dr} = \frac{1}{1 + \frac{2G_N M}{r}}. \tag{5}$$

After integration, we get

$$t(r) = r - 2G_N M \ln\left(2G_N M + r\right) + C, \tag{6}$$

where C is the integration constant. Suppose the light ray is emitted from $r = R$ at $t = 0$, then the solution becomes

$$t(r) = (r - R) + 2G_N M \ln\left(\frac{2G_N M + R}{2G_N M + r}\right). \tag{7}$$

Since r approaches infinity faster than $\ln(r)$, we see that as $r \to \infty$, $t \to \infty$.

(f) Setting $r \to \infty$, Eq. (2) becomes

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2, \tag{8}$$

which is flat spacetime.

2 Clocks in Schwarzschild

(a) We can use the Schwarzschild metric Eq. (1) to find the proper time of a static clock sitting at some $r > R_s$

$$ds^2 = -\left(1 - \frac{2G_N M}{r}\right) dt^2 = -d\tau^2, \tag{9}$$

so we get

$$d\tau = \sqrt{1 - \frac{2G_N M}{r}} dt. \tag{10}$$

We recall that t corresponds to the proper time of a clock sitting infinitely far away, so a clock sitting at r ticks faster by a factor of $\sqrt{1 - 2G_N M/r}$.

(b) Using $G_N = 6.67 \cdot 10^{-11} m^3 kg^{-1} s^{-2}$, $M = 2 \cdot 10^{30} kg$, and $r = 7 \cdot 10^8 m$, we get

$$\sqrt{1 - \frac{2G_N M}{r}} = 0.999998. \tag{11}$$

3 Falling into a Black Hole

(a) See Fig. (1). We are using the fact that the spacetime near the horizon of a black hole is just like the region of flat spacetime near an acceleration horizon. This means the trajectory of a spaceship that hovers above the horizon of the black hole at a constant radius is a hyperbola on the spacetime diagram (corresponding to a uniformly accelerating observer in flat spacetime). On the other hand, the trajectory of the free falling observer is a straight line (corresponding to an inertial observer in flat spacetime). Note that these approximations are only valid near the horizon of the black hole.

(b) See Fig. (2).

(c) People on the rocket will receive the light signal from you at increasingly longer intervals measured by a clock on the spaceship. This means they see you age more slowly. Note that the light signal that you emit at the horizon will only reach the spaceship at the infinite future. This means people on the spaceship will never see you cross the horizon. If

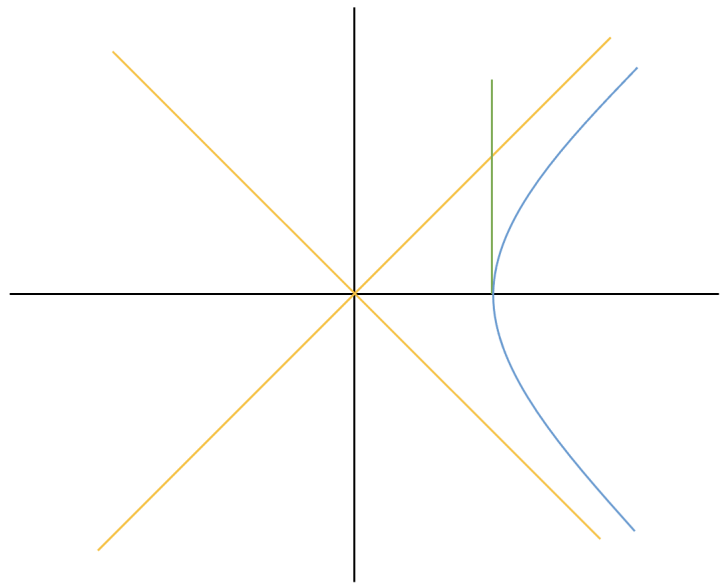

Figure 1: Green line: the trajectory of an observer free falling into the black hole. Blue hyperbola: The trajectory of a spaceship hovering over the black hole at a constant radius. The spacetime diagram is only accurate in the region near the black hole horizon.

they first sit and watch for while, it is still possible for them to come and rescue you, as long as they can reach you before you have crossed the horizon.

(d) See Fig. (3). You will receive light signals from the spaceship at increasing longer intervals as well. This means you will see that people on the spaceship age more slowly. If this is not immediately apparent to you from the spacetime diagram, think this way: the spacetime near the black hole horizon is just like the region of flat spacetime near an acceleration horizon. If you are an inertial observer, you would find the clock on an uniformly accelerating spaceship runs slowly due to time dilation. The same thing is true here. An observer free falling into a black hole would find the clock on a spaceship that is staying at a constant distance from the black hole runs slow.

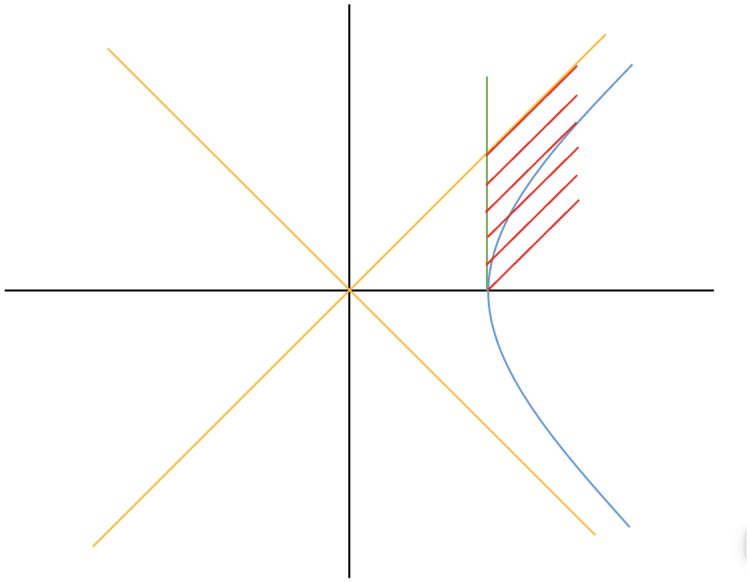

Figure 2: Same as Fig. (1) but with light signals emitted at constant time intervals from the free falling observers (red lines).

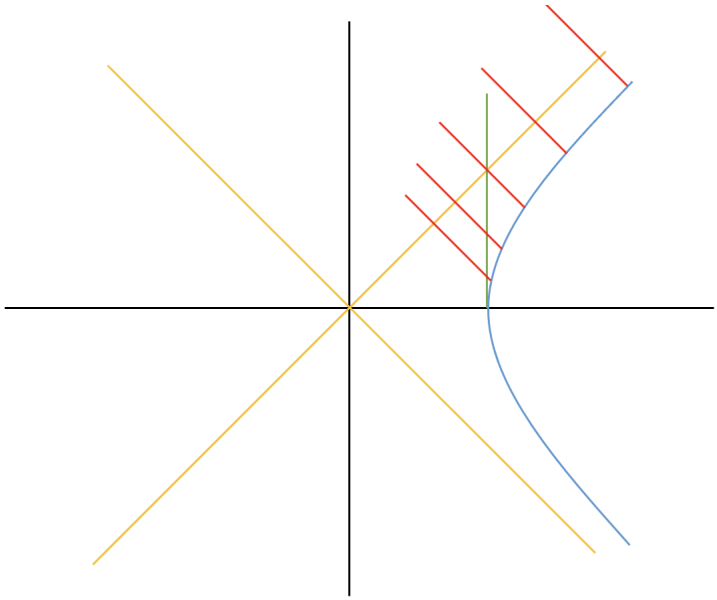

Figure 3: Now the red lines represent light signals emitted at constant time intervals from the spaceship.